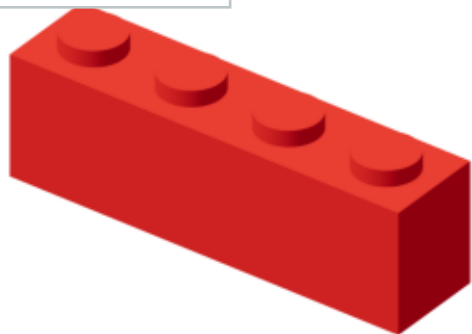


parts: INPUT

Garden Bench · easy



Isolated part

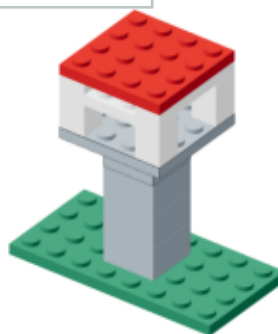
GROUND TRUTH

```
type = 3010  
color = red  
studs = 4
```

Type + color + studs

relations: INPUT

Watchtower · medium



Two queried parts

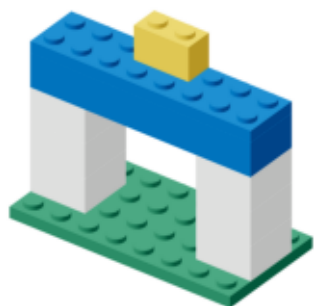
GROUND TRUTH

```
connected = true  
above = false  
studs = 4  
distance = 1
```

Typed relation tuple

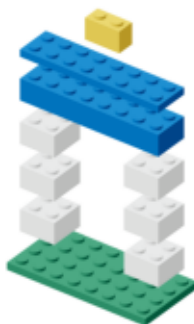
reconstruct: INPUT

Garden Gate · easy



Four assembled views + BOM

GROUND TRUTH



Full typed pose multiset

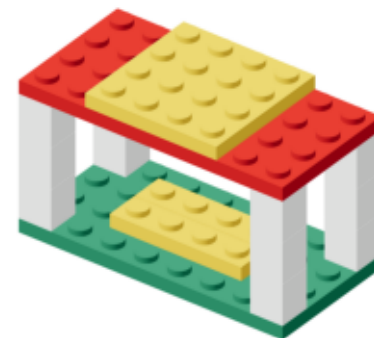
generate: INPUT

Park Pavilion · medium

```
extent = 8×4×12  
anchor = (0,0,0)  
pieces ≤ 20  
colors ≥ 4  
components = 1  
max degree ≥ 6
```

constraints only; no unique target image

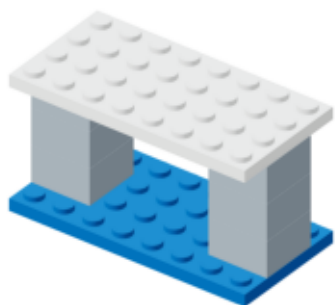
GROUND TRUTH



One satisfying witness

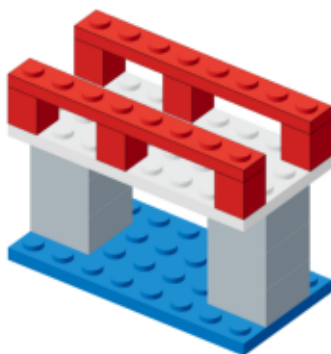
complete: INPUT

Canal Bridge · medium



Bridge missing guard rails + reference + BOM

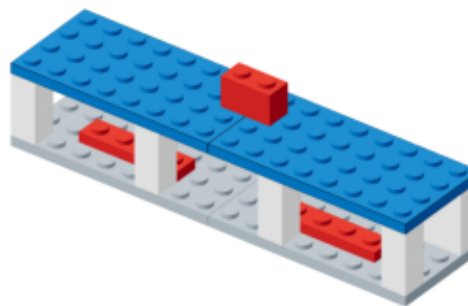
GROUND TRUTH



Guard rails restored

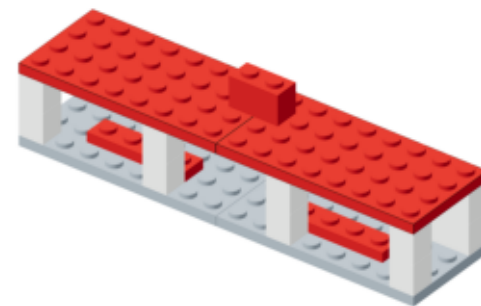
edit: INPUT

Railway Station · hard



Station + recolor command

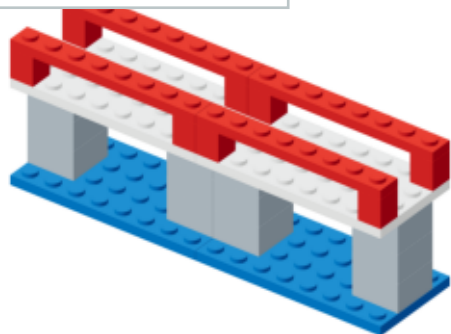
GROUND TRUTH



Red canopy, geometry preserved

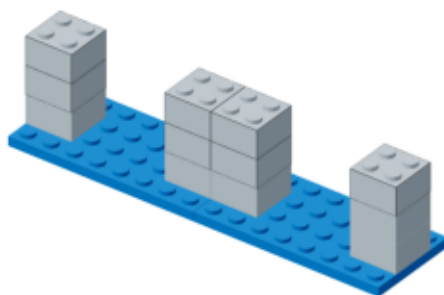
plan: INPUT

Double-Span Bridge · hard



Target geometry

GROUND TRUTH



Verified legal sequence

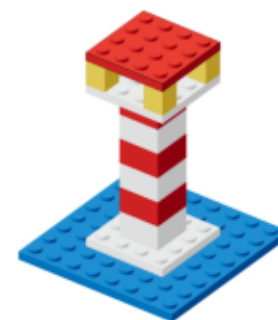
repair: INPUT

Lighthouse · hard



Faulty lighthouse + reference

GROUND TRUTH



Roof restored + fault localized